Exam: Electric and Magnetic Fields

Name: _____ Date: _____

Instructions:

- Time: 45 minutes. Allowed aids: calculator. The formulas you need are given below.
- Show your work: write down the formula, insert values with units, and give the result with a sensible number of significant figures.
- Draw a sketch wherever it helps.
- Write your name on this cover sheet only. Write your solutions in the space below each question. If you need more room, use the last page.

Formulas

Electric field: $E = \dfrac{F}{q}$ Field of a point charge: $E = k\,\dfrac{Q}{r^2}$

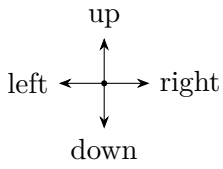
Field of a long, straight wire: $B = \dfrac{\mu_0}{2\pi}\,\dfrac{I}{r}$ Weight: $F_G = mg$

Newton's second law: $F = m\,a$

Constants

$k = \frac{1}{4\pi\varepsilon_0} = 8.99\cdot10^{9}\,\mathrm{N\,m^2/C^2}$, $\mu_0 = 4\pi\cdot10^{-7}\,\mathrm{T\,m/A}$, $e = 1.602\cdot10^{-19}\,\mathrm{C}$, $m_e = 9.109\cdot10^{-31}\,\mathrm{kg}$, $g = 9.81\,\mathrm{m/s^2}$

Directions



All directions refer to the page held upright in front of you: *up*, *down*, *left* and *right* as drawn on the page; gravity points *down*.

⊙ out of the page (towards you) ⊗ into the page (away from you)

Question:	1	2	3	4	5	6	7	Total
Points:	9	7	6	5	4	5	7	43
Score:								

Scoring (solution key)

Each **1 pt** in the solutions is one point, awarded for the step or answer written directly before it. There are no half points.

- Follow-up errors: a correct method applied to a wrong earlier result still earns its points.
- A numerical result earns its point only with a correct unit.
- Direction answers (left, right, …) need the short justification where the question asks for one.

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Do not write on this sheet: it is removed before marking.

1. Short questions.

   (a) (3 points) A test charge $q = +2.0\,\text{nC}$ placed at point P experiences a force of $6.0 \cdot 10^{-5}\,\text{N}$ to the right. (i) What is the electric field at P (magnitude and direction)? (ii) What force would a charge of $-5.0\,\text{nC}$ experience at P? (iii) What is the field at P after the test charge has been removed?

   Solution: (i) $E = F/q = 6.0 \cdot 10^{-5}/2.0 \cdot 10^{-9}\,\text{N/C} = 3.0 \cdot 10^{4}\,\text{N/C}$, to the right. **1 pt**
   (ii) $F = |q|E = 5.0 \cdot 10^{-9} \cdot 3.0 \cdot 10^{4}\,\text{N} = 1.5 \cdot 10^{-4}\,\text{N}$, to the left (negative charge). **1 pt**
   (iii) Unchanged, $3.0 \cdot 10^{4}\,\text{N/C}$ to the right: the field is created by the other charges, not by the test charge. **1 pt**

   (b) (3 points) Convert: (i) $0.45\,\text{mT}$ into µT, (ii) $3.2 \cdot 10^{5}\,\text{N/C}$ into kN/C, (iii) $1.2 \cdot 10^{4}\,\text{nC}$ into mC.

   Solution: (i) $450\,\mu\text{T}$ **1 pt** (ii) $320\,\text{kN/C}$ **1 pt** (iii) $1.2 \cdot 10^{-2}\,\text{mC}$ **1 pt**

   (c) (3 points) A loop of wire that carries a current acts as a magnet (an electromagnet). What happens to its magnetic field if (i) the current is increased, (ii) the current is reversed? (iii) Name one advantage of such an electromagnet compared to a permanent magnet.

   Solution: (i) The field becomes stronger ($B \propto I$). **1 pt**
   (ii) The field reverses its direction: the N and S poles swap. **1 pt**
   (iii) It can be switched on and off (or its strength adjusted) with the current. **1 pt**

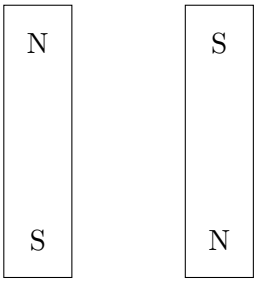
2. Field lines.

   (a) (4 points) Sketch the electric field lines (with direction) of (i) a single *negative* point charge and (ii) a positive and a negative charge of equal magnitude.

   (i) ⊖ **(ii)** ⊕ ⊖

   Solution: (i) Shape: straight radial lines. **1 pt** Direction: **towards** the charge. **1 pt**
   (ii) Shape: lines from one charge to the other, curved (densest between the charges). **1 pt**
   Direction: from + to −. **1 pt**

   (b) (3 points) Two bar magnets lie side by side, with opposite orientation (see sketch). Sketch the magnetic field lines (with direction). Where is the field strongest?



Solution: Sketch: closed loops around both magnets: from the upper N pole of the left magnet over to the upper S pole of the right magnet, and from the lower N pole of the right magnet over to the lower S pole of the left magnet (superposition of both fields). **1 pt**
Direction: outside the magnets from N to S (clockwise around the pair). **1 pt**
Strongest close to the poles, where the lines are densest (between the facing ends). **1 pt**

3. Two positive charges lie on a line: $Q_1 = +9.0\,\mu\text{C}$ at $x = 0$ and $Q_2 = +1.0\,\mu\text{C}$ at $x = 20\,\text{cm}$.
   (a) (3 points) Find the magnitude and direction of the electric field at the midpoint M ($x = 10\,\text{cm}$).

   Solution: $E_1 = k\dfrac{Q_1}{r^2} = 8.99 \cdot 10^9 \cdot \dfrac{9.0 \cdot 10^{-6}}{0.10^2}\,\text{N/C} \approx 8.09 \cdot 10^6\,\text{N/C}$ (to the right, away from Q_1). **1 pt**
   $E_2 = 8.99 \cdot 10^9 \cdot \dfrac{1.0 \cdot 10^{-6}}{0.10^2}\,\text{N/C} \approx 8.99 \cdot 10^5\,\text{N/C}$ (to the left, away from Q_2). **1 pt**
   $E = E_1 - E_2 \approx 7.2 \cdot 10^6\,\text{N/C}$, to the **right**. **1 pt**

   (b) (3 points) At which point on the x-axis is the electric field zero?

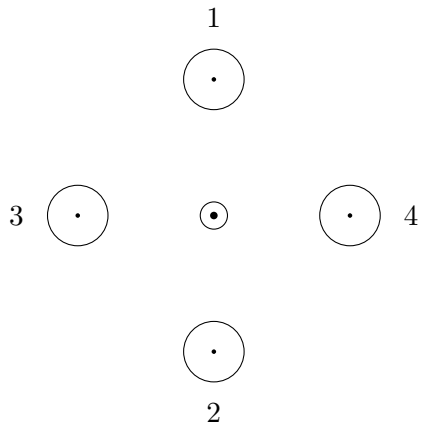
   Solution: Only between the charges do the two fields point in opposite directions. **1 pt**
   $k\dfrac{Q_1}{x^2} = k\dfrac{Q_2}{(0.20 - x)^2} \Rightarrow \dfrac{0.20 - x}{x} = \sqrt{\dfrac{Q_2}{Q_1}} = \dfrac{1}{3}$ **1 pt**
   $\Rightarrow x = 0.15\,\text{m} = 15\,\text{cm}$ (i.e. 5 cm from Q_2). **1 pt**

4. (5 points) An electron is in a uniform electric field of $500\,\text{N/C}$ that points up. Calculate the electric force on the electron (magnitude and direction) and compare it with its weight. What is the acceleration of the electron?

   Solution: $F = eE = 1.602 \cdot 10^{-19}\,\text{C} \cdot 500\,\text{N/C} \approx 8.0 \cdot 10^{-17}\,\text{N}$ **1 pt**, pointing **down** (negative charge). **1 pt**
   Weight: $F_G = m_e g = 9.109 \cdot 10^{-31} \cdot 9.81\,\text{N} \approx 8.9 \cdot 10^{-30}\,\text{N}$. **1 pt**
   The electric force is about 10^{13} times larger, so the weight is negligible. **1 pt**
   $a = F/m_e = 8.0 \cdot 10^{-17}/9.109 \cdot 10^{-31}\,\text{m/s}^2 \approx 8.8 \cdot 10^{13}\,\text{m/s}^2$ (downward). **1 pt**

5. (4 points) **Compasses around a wire.** A long, straight wire is perpendicular to the page and carries a current *out of* the page. Four compasses are placed around it (neglect the Earth's

magnetic field). Draw the direction in which the N pole of each compass points. What changes if the current is reversed?

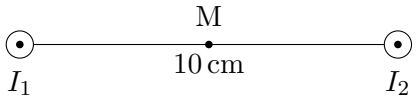


Solution: The field lines are circles around the wire; each needle points along the field. **1 pt**
Right-hand rule (thumb out of the page): the circles run counterclockwise. Two points if all four directions are correct, one point for two or three:
1 (above): **left**. 2 (below): **right**. 3 (left): **down**. 4 (right): **up**. **1 pt** **1 pt**
Reversed current: the field reverses, all needles turn by 180°. **1 pt**

6. (5 points) **Power line.** A long, horizontal power line in the plane of the page carries a current of 200 A to the right. (a) At what distance from the line is its magnetic field as strong as the Earth's magnetic field ($\approx 50\,\mu$T)? (b) In which direction does the line's field point at a point below the line?

Solution: (a) $B = \dfrac{\mu_0}{2\pi}\dfrac{I}{r}$ **1 pt** $\Rightarrow r = \dfrac{\mu_0}{2\pi}\dfrac{I}{B}$ **1 pt** $= 2 \cdot 10^{-7} \cdot \dfrac{200}{5.0 \cdot 10^{-5}}\,\text{m} = 0.80\,\text{m}.$ **1 pt**
(b) Right-hand rule: thumb to the right, fingers curl around the line. **1 pt**
Below the line the field points **into the page**. **1 pt**

7. Two long, straight wires are perpendicular to the page and 10 cm apart. Both currents point *out of* the page: $I_1 = 30$ A in the left wire, $I_2 = 10$ A in the right wire.



(a) (4 points) Find the magnitude and direction of the magnetic field at the midpoint M.

Solution: $B_1 = 2 \cdot 10^{-7} \cdot \dfrac{30}{0.050}\,\text{T} = 1.2 \cdot 10^{-4}\,\text{T}.$ **1 pt** $B_2 = 2 \cdot 10^{-7} \cdot \dfrac{10}{0.050}\,\text{T} = 4.0 \cdot 10^{-5}\,\text{T}.$ **1 pt**
Both fields circle counterclockwise: at M, B_1 points up, B_2 down. **1 pt**
$B = B_1 - B_2 = 8.0 \cdot 10^{-5}\,\text{T}$, pointing **up**. **1 pt**

(b) (3 points) At which point between the wires is the magnetic field zero?

Solution: Between the wires the two fields point in opposite directions; they cancel where they are equally strong: **1 pt** $\dfrac{I_1}{x} = \dfrac{I_2}{0.10 - x}$ **1 pt**
$\Rightarrow 30(0.10 - x) = 10x \Rightarrow x = 0.075\,\text{m} = 7.5\,\text{cm}$ from the left wire. **1 pt**